

What a VIP package is

Module eleven mapped classic testbench habits onto UVM roles

Starter and incomplete presets

- Starter loads a complete UART VIP
- Scenario presets show what breaks handoff
- Missing checker keeps agent, coverage, and docs but drops protocol rules, INCOMPLETE
- Missing coverage drops bins while stimulus and checks remain, INCOMPLETE
- Missing docs drops the self-test handoff even when code pieces exist, INCOMPLETE
- Agent only is useful scaffolding, not a VIP yet

Browser lab

Digital Design and Verification Platform

HomeTools

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INTERACTIVE TOOL

VIP anatomy

Sketch a reusable **verification IP package**: **agent** + **checker** + **coverage** + **docs**.
Starter: UART VIP with all four pieces present — complete.

Starter example: UART VIP with agent + checker + coverage + docs — package COMPLETE.

Load starter example

Challenges 0 / 22 cleared

Quiz: VIP: A verification IP (VIP) package is mainly...

☐ a reusable agent + checker + coverage + docs handoff

☐ only a Makefile PHONY target

☐ a replacement for the DUT

☐ just +UVM_TESTNAME

Show hint

Check

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Core ideas

agent

Drive / observe one protocol interface.

checker

Passive bus-rule watch shipped with the VIP.

coverage

Bins that prove which scenarios ran.

docs

API + examples + self-test for handoff.

Browser lab starter

Real RTL/TB practice

- In Track A, restate the four VIP deliverables in one sentence each
- Docs should stay blank
- Optional: peek at the linked UART RTL or TB sketch and label which folders would hold
- Remember: scoreboard payload compare often lives in the environment
- Your job is to read a block diagram and know what a consumer still has to wire in the env

Pitfalls to watch

- Do not call an agent folder a finished VIP, handoff needs checker, coverage, and docs too
- Do not confuse checker with scoreboard, the checker flags protocol legality
- Coverage is not optional decoration, it proves which bins your VIP scenarios actually hit
- Docs are not README fluff, they include a self-test so integration does not stall
- And remember: this lab teaches package anatomy
- Factory overrides, RAL, and full UVM env wiring come in later courses, not here

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load starter, then Demo incomplete and name the missing piece
- On paper, draw `uart_vip` with four subfolders and mark which are empty for agent-only
- When you are ready